

TF-OVCOS: A Four-Part Evaluation Framework for Training-Free Open-Vocabulary Camouflaged Object Segmentation

Proposal Draft v19 – concrete method roster and E3 dataset protocol

Draft for Discussion

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1 Core Idea

This proposal studies **training-free open-vocabulary camouflaged object segmentation** (TF-OVCOS). Given an image I and a candidate vocabulary $\mathcal{C}$, each method must output one mask–category pair:

$$I + \mathcal{C} \rightarrow (\hat{M}, \hat{y}), \quad \hat{y} \in \mathcal{C}. \quad (1)$$

The main goal is not to introduce another trained OVCOS model. Instead, we define a concrete and reproducible evaluation framework for frozen public foundation-model pipelines.

Strict setting. The main rows are **training-free and MLLM/API-free**: no OV-Camo/COS training, no prompt tuning, no adapter training, no validation threshold tuning, no inference-time generative MLLM calls, and no closed-source API-VLMs. Frozen public models such as CLIP, SigLIP, GroundingDINO, DINOv2, SAM, and SAM2 are allowed.

Compact trained references. Following advisor feedback, training-based rows are kept intentionally small. The main paper keeps only OVCoser [1] and SuCLIP [2] as compact OVCOS references. They are not strict-TF competitors.

2 Evaluation Protocol

Instead of designing multiple small OVCamo vocabulary variants, we organize the paper around four evaluation questions: standard effectiveness, category ambiguity, source-to-target generalization, and efficiency.

Why this replaces the previous S1/S2/S3 structure. The previous scenario design became too vocabulary-centric. The revised E1–E4 framework makes each evaluation answer a different question: E1 asks whether the method works on the official OVCOS task, E2 asks how label ambiguity appears after successful localization, E3 asks how methods transfer across datasets/domains, and E4 asks how much each method costs.

External-target rule. For external mask-only targets, the input vocabulary for text-conditioned methods is fixed to the official OVCamo full vocabulary $\mathcal{C}_{75} = \mathcal{C}_{seen}^{14} \cup \mathcal{C}_{unseen}^{61}$. The predicted label is ignored and only the predicted mask is scored. This avoids target-specific prompt engineering or hand-designed external category vocabularies. Since OVCamo is assembled from existing camouflaged-scene data, external E3 targets are deduplicated against

Table 1: Final E1–E4 evaluation protocol. E1 is the primary leaderboard; E2–E4 explain ambiguity, cross-dataset generalization, and cost.

Evaluation	Data / input	Main output and metrics	Role
E1 Standard OVCOS effectiveness	OVCamo-TE image + official 61 unseen OVCamo labels.	One mask–category pair. Primary metric is official $cIoU$; also report cS_m , cF_β^ω , cE_m , $cMAE$, and $cBIoU$.	Primary strict-TF leaderboard.
E2 Category ambiguity robustness	Same E1 predictions and official OVCamo labels. No external siblings or ontology-expanded labels.	Loc@0.5, Exact@0.5, ClsErr@Loc, top official-label confusion pairs, and optional ambiguity graph.	Explain whether failures are localization errors or label ambiguity.
E3 Domain / cross-dataset generalization	Strict-TF methods use no source training. Trained references use OVCamo-TR/base. Targets are OVCamo-TE, CAMO-TE, COD10K-TE-Camo, and NC4K.	Class-aware metrics on OVCamo-TE; mask-only IoU, S_m , F_β^ω , E_m , MAE on external targets.	Compare training-free transfer with a small trained-reference set.
E4 Efficiency comparison	Same method rows, fixed hardware, fixed input size, fixed adapters.	Training time, test time per image, peak memory, trainable parameters, number of model calls, and module requirements.	Expose cost behind performance.

Table 2: Concrete E3 source-to-target dataset plan. External camouflage targets are used for mask-only generalization unless reliable class labels can be mapped to an OVCOS vocabulary.

Dataset / split	Role in E3	What it provides	Metrics used	Main caveat
OVCamo-TR/base [1]	Source for trained references	Official OVCamo training/base split used only by trained references.	Training only; never used for strict-TF tuning.	Trained rows only.
OVCamo-TE [1]	Class-aware target	Camouflaged images, masks, class labels, and official 61 unseen labels.	$cIoU$, cS_m , cF_β^ω , cE_m , $cMAE$, $cBIoU$.	Same target as E1.
CAMO-TE [24]	External mask-only target	250 camouflaged-object test images with objectness masks.	IoU , S_m , F_β^ω , E_m , MAE.	No official OVCOS vocabulary.
COD10K-TE-Camo [25]	External mask-only target	2,026 camouflaged-object test images from COD10K-TE.	IoU , S_m , F_β^ω , E_m , MAE.	Use only camouflaged subset.
NC4K [26]	External mask-only target	4,121 camouflaged-object test images.	IoU , S_m , F_β^ω , E_m , MAE.	Deduplicate against OV-Camo.

OVCamo by filename/MD5 matching and perceptual hashing before reporting. CHAMELEON is excluded from the main E3 table because it has only 76 images and is better suited for appendix sanity checks.

3 Metrics

E1 main metrics. The primary ranking metric is official $cIoU$. The main table reports $cIoU$, cS_m , cF_β^ω , cE_m , $cMAE$, and $cBIoU$. The full official metric suite, including cF_β , is kept in the appendix or spreadsheet.

E2 mask-wise ambiguity metrics. Following the mask-wise motivation of recent open-vocabulary evaluation work [3], E2 uses only official OVCamo labels and model predictions:

$$\text{Loc@0.5} = \frac{1}{N} \sum_i \mathbf{1} \left[\text{IoU}(\hat{M}_i, G_i) \geq 0.5 \right], \quad (2)$$

$$\text{Exact@0.5} = \frac{1}{N} \sum_i \mathbf{1} \left[\text{IoU}(\hat{M}_i, G_i) \geq 0.5 \right] \mathbf{1}[\hat{y}_i = y_i], \quad (3)$$

$$\text{ClsErr@Loc} = 1 - \frac{\text{Exact@0.5}}{\max(\text{Loc@0.5}, \epsilon)}. \quad (4)$$

For localized but misclassified predictions, we count directed official-label confusion edges $y_i \rightarrow \hat{y}_i$ and report the most frequent pairs.

E3 generalization metrics. For OVCamo-TE, E3 reuses E1 class-aware metrics. For CAMO-TE, COD10K-TE-Camo, and NC4K, it reports mask-only IoU, S_m , F_β^ω , E_m , and MAE. We also report an external-domain average and, when source/in-domain performance exists, a target drop:

$$\Delta_{\text{gen}} = \text{Score}_{\text{OVCamo-TE}} - \text{Score}_{\text{external avg}}. \quad (5)$$

E4 cost metrics. E4 reports training time, inference time per image, peak GPU memory, trainable parameters, model calls, and module requirements.

4 Methods and Adapter

The benchmark uses explicit method rows rather than only family names. Final inclusion still requires a code/weight check, but the proposal fixes the intended roster below.

Adapter rule. Every method is converted to $(\hat{M}, \hat{y})$ using pre-declared thresholds, prompts, resize sizes, SAM settings, and post-processing. Constants are fixed before OVCamo-TE evaluation and are not tuned per method. MCC re-ranking is shown as an ablation row under proposal-based pipelines rather than as a new method family.

Table 3: Concrete method roster and OVCOS adapters. Rows marked backup are included if public code/weights are stable enough; otherwise they move to appendix with an implementation note.

Family	E1/E2 rows	Default E3 rows	Unified adapter	OVCOS
CLIP-only attention-edited OVSS	/ MaskCLIP [11], CLIP-DIY [12], SCLIP [13], MaskCLIP, SCLIP, NACLIP [14]; backup: CLIPtrase [15], SC-CLIP [16], ResCLIP [17]	MaskCLIP, SCLIP, NACLIP, ResCLIP	Convert class score maps into candidate masks by fixed thresholding / connected components or SAM refinement, then select one pair.	
CLIP + VFM OVSS	ProxyCLIP [18], CorrCLIP [19], Trident [20]; backup: CASS [21]	ProxyCLIP, CorrCLIP, Trident	Use VFM-refined patch correlations or SAM/DINO spatial priors before pair selection.	
Diffusion reference-based OVSS	/ OVDiff [22], FreeDA [23]	FreeDA; OVDiff if code/runtime is feasible	Use diffusion-generated or diffusion-augmented visual prototypes to match class-agnostic regions with text classes.	
OV detection SAM	+ GroundingDINO [10] + SAM [8]; GroundingDINO + SAM2 [9]	GroundingDINO + SAM2	Prompt the detector with class names, convert boxes into masks with SAM/SAM2, and score mask-category pairs.	
Class-agnostic proposal + VLM naming	SAM-AMG + CLIP/SigLIP [5, 6]; DINOv2 [7] + SAM + CLIP/SigLIP; MCC variants	SAM-AMG + SigLIP; DINOv2 + SAM + SigLIP; DINOv2 + SAM + CLIP + MCC	Generate class-agnostic proposals, name proposals with a frozen VLM, and optionally re-rank by mask-category consistency.	
Compact references	trained OVCoser [1], SuCLIP [2] only	OVCoser, SuCLIP	Not strict-TF. Used as compact context for E1/E3/E4, not as primary benchmark rows.	

5 Main Tables

5.1 E1: Standard OVCOS Effectiveness

Table 4: E1 standard OVCOS effectiveness leaderboard. All strict-TF rows use OVCamo-TE and the official 61 unseen OVCamo labels. Numeric entries are filled after running the fixed adapters. Compact trained references are shown only for context and are not strict-TF competitors.

Method	Adapt.	Strict TF?	$cIoU\uparrow$	$cS_m\uparrow$	$cF_{\beta}^{\omega}\uparrow$	$cE_m\uparrow$	$cMAE\downarrow$	$cBioU\uparrow$
CLIP-only / attention-edited OVSS								
MaskCLIP	CM	Yes						
CLIP-DIY	CM	Yes						
SCLIP	CM	Yes						
NACLIP	CM	Yes						
CLIPtrase	CM	Yes						
SC-CLIP	CM	Yes						
ResCLIP	CM	Yes						
CLIP + VFM OVSS								
ProxyCLIP	VM	Yes						
CorrCLIP	VM	Yes						
Trident	VM	Yes						
CASS	VM	Yes						
Diffusion / reference-based OVSS								
OVDiff	DM	Yes						
FreeDA	DM	Yes						
OV detection + SAM								
GroundingDINO + SAM	DS	Yes						
GroundingDINO + SAM2	DS	Yes						
Class-agnostic proposal + VLM naming								
SAM-AMG + CLIP	PN	Yes						
SAM-AMG + SigLIP	PN	Yes						
DINOv2 + SAM + CLIP	PN	Yes						
DINOv2 + SAM + SigLIP	PN	Yes						
DINOv2 + SAM + CLIP + MCC	PN	Yes						
Compact training-based references (not strict TF)								
OVCoser	Trained	No	0.443	0.579	0.490	0.616	0.336	–
SuCLIP	Trained	No	0.540	0.667	0.594	0.722	0.242	–

5.2 E2: Category Ambiguity Robustness

Table 5: E2 mask-wise category ambiguity analysis. The table uses the same predictions as E1 and only official OVCamo labels. A confusion edge $y \rightarrow \hat{y}$ is recorded when the predicted mask is localized but the category is not exact.

Method	Loc@0.5↑	Exact@0.5↑	ClsErr@Loc↓	#Edges	Top official-label confusions
MaskCLIP					
CLIP-DIY					
SCLIP					
NACLIP					
CLIPtrase					
SC-CLIP					
ResCLIP					
ProxyCLIP					
CorrCLIP					
Trident					
CASS					
OVDiff					
FreeDA					
GroundingDINO + SAM					
GroundingDINO + SAM2					
SAM-AMG + CLIP					
SAM-AMG + SigLIP					
DINOv2 + SAM + CLIP					
DINOv2 + SAM + SigLIP					
DINOv2 + SAM + CLIP + MCC					
OVCoser (ref.)					
SuCLIP (ref.)					

5.3 E3: Domain / Cross-Dataset Generalization

Table 6: E3 source-to-target generalization table. OVCamo-TE remains class-aware; CAMO-TE, COD10K-TE-Camo, and NC4K are external mask-only targets. Strict-TF rows use no source training. Trained references are limited to OVCoser and SuCLIP.

Method	Source / training data	OVCamo-TE <i>cIoU</i> ↑	CAMO-TE mask↑	COD10K-TE-Camo mask↑	NC4K mask↑	Ext. avg / Δ_{gen}
Strict training-free rows						
MaskCLIP	None					
CLIP-DIY	None					
SCLIP	None					
NACLIP	None					
CLIPtrase	None					
SC-CLIP	None					
ResCLIP	None					
ProxyCLIP	None					
CorrCLIP	None					
Trident	None					
CASS	None					
OVDiff	None; offline diffusion references only					
FreeDA	None; offline diffusion prototypes only					
GroundingDINO + SAM	None					
GroundingDINO + SAM2	None					
SAM-AMG + CLIP	None					
SAM-AMG + SigLIP	None					
DINOv2 + SAM + CLIP	None					
DINOv2 + SAM + SigLIP	None					
DINOv2 + SAM + CLIP + MCC	None; post-hoc re-ranking only					
Compact trained references						
OVCoser	OVCamo-TR/base	0.443				
SuCLIP	OVCamo-TR/base + prompt/alignment training	0.540				

E3 reporting rule. For external targets, every text-conditioned method receives the same fixed OVCamo-75 vocabulary. The final class label is ignored because CAMO, COD10K-TE-

Camo, and NC4K do not provide the same OVCOS category vocabulary. We therefore report mask-only scores on external targets and reserve class-aware scores for OVCamo-TE.

5.4 E4: Efficiency Comparison

Table 7: E4 efficiency comparison. Exact numbers are filled after running every method on the same GPU, input size, and adapter settings. Training-based methods are intentionally limited to two compact references.

Method	Training?	Train time	Test cost	Major modules / notes
MaskCLIP	No	0	report	CLIP dense label extraction baseline.
CLIP-DIY	No	0	report	CLIP patch inference plus unsupervised localization prior.
SCLIP	No	0	report	CLIP with correlative self-attention.
NACLIP	No	0	report	CLIP neighbour-aware attention adaptation.
CLIPtrase	No	0	report	CLIP self-attention / dense inference modification.
SC-CLIP	No	0	report	Self-calibrated CLIP; no new trainable parameters.
ResCLIP	No	0	report	Residual cross-correlation attention and semantic feedback refinement.
ProxyCLIP	No	0	report	CLIP plus VFM proxy attention.
CorrCLIP	No	0	report	CLIP with SAM/self-supervised patch correlation reconstruction.
Trident	No	0	report	CLIP + DINO + SAM high-resolution training-free pipeline.
CASS	No	0	report	CLIP with VFM spectral object-context distillation at inference.
OVDiff	No	offline synth.	report	Diffusion-generated support/prototype stage; no OVCamo training.
FreeDA	No	offline proto.	report	Offline diffusion-augmented prototype generation.
GroundingDINO + SAM	No	0	report	Open-set detector plus SAM mask generation.
GroundingDINO + SAM2	No	0	report	Open-set detector plus SAM2 mask generation.
SAM-AMG + CLIP	No	0	report	Class-agnostic SAM proposals plus CLIP naming.
SAM-AMG + SigLIP	No	0	report	Class-agnostic SAM proposals plus SigLIP naming.
DINOv2 + SAM + CLIP	No	0	report	DINOv2-guided proposals plus CLIP naming.
DINOv2 + SAM + SigLIP	No	0	report	DINOv2-guided proposals plus SigLIP naming.
DINOv2 + SAM + CLIP + MCC	No	0	report	Adds post-hoc mask-category consistency scoring.
OVCoser	Yes	report	report	Compact trained OVCOS reference.
SuCLIP	Yes	report	report	Compact trained / prompt-adapted OVCOS reference.

5.5 E1 Ablations

Table 8: E1 ablations. These explain failures in the standard OVCOS leaderboard but are not separate evaluations.

Ablation	Setting	Top-1↑	Top-5↑	IoU↑	BLoU↑
GT-mask recognition	CLIP/SigLIP on GT foreground mask; localization removed.			–	–
GT-mask recognition	Alpha-mask naming with full context; localization removed.			–	–
GT-label localization	GroundingDINO + SAM/SAM2 with GT class prompt; recognition removed.	–	–		
GT-label localization	Dense CLIP/VFM map for GT class + SAM; recognition removed.	–	–		
Proposal oracle	Best SAM-AMG/DINOv2 proposal by GT IoU; naming removed.	–	–		

A Appendix Tables

Table A1: Full official OVCOS class-aware metric suite for every strict-TF method.

Method	$cS_m \uparrow$	$cF_\beta^\omega \uparrow$	$cMAE \downarrow$	$cF_\beta \uparrow$	$cE_m \uparrow$	$cIoU \uparrow$
Strict TF rows, same E1 setting as Table 4						
MaskCLIP						
CLIP-DIY						
SCLIP						
NACLIP						
CLIPtrase						
SC-CLIP						
ResCLIP						
ProxyCLIP						
CorrCLIP						
Trident						
CASS						
OVDiff						
FreeDA						
GroundingDINO + SAM						
GroundingDINO + SAM2						
SAM-AMG + CLIP						
SAM-AMG + SigLIP						
DINOv2 + SAM + CLIP						
DINOv2 + SAM + SigLIP						
DINOv2 + SAM + CLIP + MCC						
Compact training-based references						
OVCoser	0.579	0.490	0.336	0.520	0.616	0.443
SuCLIP	0.667	0.594	0.242	0.633	0.722	0.540

Table A2: CamoPrompt ablation. These rows are non-strict references and excluded from E1 ranking.

Base method	Prompt template	$cIoU \uparrow$	$cS_m \uparrow$	$cF_\beta^\omega \uparrow$	$cBioU \uparrow$	Note
CLIP heatmap + SAM	“a camouflaged {class}”					Non-strict prompt ablation
GroundingDINO + SAM	“a camouflaged {class}”					Non-strict prompt ablation
SAM-AMG + CLIP naming	camouflage-aware template set					Non-strict prompt ablation
DINOv2 + SAM + CLIP	camouflage-aware template set					Non-strict prompt ablation

Table A3: E2 ambiguity graph summary. Filled after collecting localized-but-misclassified pairs.

Method group	#Matched pairs	#Mismatch edges	Top community size	Example official-label communities
Dense-map TF methods				
Detector+SAM TF methods				
Proposal+naming TF methods				
MCC variants				

Table A4: Per-method exclusion grounds for methods outside the strict TF leaderboard.

Method	Task training?	Prompt/adapt. training?	MLLM call?	API-VLM?	Exclusion reason
OVCoser	Yes	Yes	No	No	Kept only as compact trained reference.
SuCLIP	Yes	Yes	No	No	Kept only as compact trained reference.
COCUS / cascaded VLM OVCOS	Yes	Yes	No	No	Related work or optional appendix; excluded from main reference set to keep training rows small.
DSS / Discover-Segment-Select	No	No	Yes	No	Uses inference-time MLLM mask selection; related zero-shot COS only.
GenSAM / ProMaC	No	No	Yes	No	Inference-time generative MLLM; related work only.
API-VLM + SAM	No	No	Depends	Yes	Closed-source API; not reproducible under strict protocol.

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